Structural Change

Lecture 6 Cheatsheet — Macroeconomics I (BSE)

Exam Reference

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1 Notation and Definitions

Sector Variables

A = Agriculture, M = Manufacturing, S = Services

c_t^i = per-capita consumption in sector $i \in \{A, M, S\}$

L_{it} = employment in sector i

Y_{it} = output in sector i

p_i = price of good i

I = aggregate income/expenditure

γ^A, γ^S = Stone-Geary subsistence parameters($\gamma^A, \gamma^S > 0$)

η_A, η_M, η_S = expenditure-share parameters, $\eta_A + \eta_M + \eta_S = 1$

g = aggregate consumption growth rate (BGP)

n = population growth rate

B_i = sector-i productivity level

A_{it} = TFP of sector i

$a_i = \dot{A}_{it}/A_{it}$ = TFP growth rate in sector i

ε = elasticity of substitution between goods (CES aggregator)

α_i = capital share of sector i

$k_t = K_{1t}/K_t$ = capital allocation fraction

$\lambda_t = L_{1t}/L_t$ = labor allocation fraction (sector 1)

σ = intertemporal elasticity of substitution (HRV preferences)

$\bar{c}_i$ = non-homothetic threshold parameter (HRV)

Key Notation Conventions

- Dot notation: $\dot{x}_t \equiv dx_t/dt$, growth rate $\gamma_x \equiv \dot{x}/x$.
- Subscript t on time-varying variables may be dropped when clear from context.
- "Sector 1" and "Sector 2" used in the supply model; agriculture/manufacturing/services in demand model.
- Engel curve: relationship between income and quantity demanded holding prices fixed.

2 Stylized Facts of Structural Change

Cross-Country Patterns (as Income Rises)

Sector	Employment Share	Output Share	Pattern
Agriculture	L_A/L	Y_A/Y	Declines monotonically
Manufacturing	L_M/L	Y_M/Y	Inverse-U (rises then falls)
Services	L_S/L	Y_S/Y	Rises monotonically

Productivity Growth Rates (Duarte & Restuccia, 1956–2004)

Sector	Average TFP Growth Rate
Agriculture	3.8% per year
Manufacturing	2.4% per year
Services	1.3% per year

Key observation: TFP growth rates differ systematically across sectors. Agriculture and manufacturing have high productivity growth; services have low productivity growth. Relative price changes (services become relatively expensive) and relative income effects both drive reallocation of labor.

Two Candidate Mechanisms

1. **Demand-side (non-homotheticities):** As income grows, income elasticities differ across sectors. People spend a declining share on food (necessity) and a rising share on services (luxury). Even at constant relative prices, sector shares change.

2. **Supply-side (Baumol cost disease):** Productivity growth differs across sectors. Fast-growing sectors have falling relative prices, inducing substitution. If goods are complements ($\varepsilon < 1$), the slow-growing sector absorbs factors.

3 Demand-Side: Homotheticity and Engel Curves

Homothetic Preferences

Preferences are **homothetic** if the utility function is homogeneous of degree 1:

$$U(\lambda c^A, \lambda c^M, \lambda c^S) = \lambda\, U(c^A, c^M, c^S) \quad \forall\, \lambda > 0$$

Implication for Engel curves: With homothetic preferences, all Engel curves are linear *and pass through the origin.* Doubling income exactly doubles demand for every good. Sectoral budget shares are independent of income.

Engel Curves

The Engel curve for good i is $x_i = x_i(I, p_i)$ holding prices fixed. Classification:

Type	**Income Elasticity** ε_I	**Engel Curve Shape**
Normal good	$\varepsilon_I > 0$	Increasing in I
Necessity	$0 < \varepsilon_I < 1$	Increasing and concave in I
Luxury	$\varepsilon_I > 1$	Increasing and convex in I
Inferior good	$\varepsilon_I < 0$	Decreasing in I
Homothetic	$\varepsilon_I = 1$	Linear through origin

Engel's Law

As per capita income rises, the *budget share* of food (agriculture) declines. This is the oldest stylized fact in economics (Ernst Engel, 1857). Formally: food has income elasticity less than 1, so $\partial(p_A c^A/I)/\partial I < 0$.

Non-homotheticity is required for structural change via demand. If preferences were homothetic, income growth would not change sectoral budget shares, and no demand-driven reallocation would occur. The key is that agriculture is a necessity ($\varepsilon_I^A < 1$) and services tend to be a luxury ($\varepsilon_I^S > 1$).

4 Stone-Geary Preferences and Demand Derivation

Stone-Geary Utility

The Stone-Geary utility function introduces subsistence levels:

$$u_t = (c_t^A - \gamma^A)^{\eta_A} \cdot (c_t^M)^{\eta_M} \cdot (c_t^S + \gamma^S)^{\eta_S} \tag{Stone-Geary}$$

where $\gamma^A > 0$ (subsistence floor for agriculture), $\gamma^S > 0$ (satiation level for services with reversed sign), and $\sum_i \eta_i = 1$, $\eta_i > 0$.

General compact form:

$$u = \prod_{i \in \{A,M,S\}} (c_i - \gamma_i)^{\eta_i}$$

with $\gamma^A > 0$, $\gamma^M = 0$, $\gamma^S < 0$ (so $c^S + \gamma^S$ means $c^S - |\gamma^S|$, i.e., services is a luxury).

Demand Derivation

The household maximizes u_t subject to the budget constraint $\sum_i p_i c_i = I$.

Step 1. Form the Lagrangian:

$$\mathcal{L} = \prod_i (c_i - \gamma_i)^{\eta_i} - \lambda \left(\sum_i p_i c_i - I \right)$$

Step 2. Take the FOC for good j:

$$\frac{\partial \mathcal{L}}{\partial c_j} = \eta_j (c_j - \gamma_j)^{-1} \cdot \prod_i (c_i - \gamma_i)^{\eta_i} = \lambda p_j$$

which simplifies to:

$$\frac{\eta_j \, u}{c_j - \gamma_j} = \lambda p_j \tag{FOC-j}$$

Step 3. Take the ratio of FOC-j and FOC-i:

$$\frac{\eta_j}{\eta_i} \cdot \frac{c_i - \gamma_i}{c_j - \gamma_j} = \frac{p_j}{p_i} \implies (c_i - \gamma_i) = \frac{\eta_i \, p_j}{\eta_j \, p_i} (c_j - \gamma_j)$$

Step 4. Substitute into the budget constraint $\sum_i p_i c_i = I$:

$$\sum_i p_i \gamma_i + \sum_i p_i (c_i - \gamma_i) = I$$

Using Step 3, $p_i(c_i - \gamma_i) = \eta_i \cdot (c_j - \gamma_j) \cdot p_j / \eta_j$ for all i. Summing:

$$\sum_i p_i (c_i - \gamma_i) = \frac{p_j (c_j - \gamma_j)}{\eta_j} \sum_i \eta_i = \frac{p_j (c_j - \gamma_j)}{\eta_j}$$

Therefore:

$$\sum_i p_i \gamma_i + \frac{p_j (c_j - \gamma_j)}{\eta_j} = I \implies c_j - \gamma_j = \frac{\eta_j (I - \sum_i p_i \gamma_i)}{p_j}$$

Step 5. Final demand schedule:

$$\boxed{c_j = \gamma_j + \frac{\eta_j \left(I - \sum_i p_i \gamma_i \right)}{p_j}} \tag{Demand}$$

Properties of Stone-Geary Demand

- **Linear Engel curves** but *not* through the origin (non-homothetic).
- **Agriculture ($\gamma^A > 0$):** $c^A - \gamma^A$ grows proportionally with supernumerary income. The Engel curve is concave $\Rightarrow$ necessity. Income elasticity $\varepsilon_I^A = \eta_A I/(p_A c^A) < 1$ since $p_A c^A > p_A \gamma^A + \eta_A (I - \sum p_i \gamma_i) \cdot (p_A/p_A) = \eta_A I + (1 - \eta_A) p_A \gamma^A$ ... simplifying: as $I \to \infty$, $\varepsilon_I^A \to \eta_A < 1$.
- **Manufacturing ($\gamma^M = 0$):** Engel curve passes through origin. c^M grows proportionally $\Rightarrow$ income elasticity $\to \eta_M$ (unit elastic asymptotically).
- **Services ($\gamma^S < 0$, i.e., $-|\gamma^S|$):** $c^S + |\gamma^S|$ grows proportionally. The Engel curve is convex $\Rightarrow$ luxury. As $I \to \infty$, $\varepsilon_I^S \to \eta_S < 1$ but for moderate I the elasticity exceeds 1.
- **Budget share of agriculture:** $p_A c^A / I \to \eta_A$ as $I \to \infty$ and $> \eta_A$ for finite I.
- **Budget share of services:** $p_S c^S / I \to \eta_S$ as $I \to \infty$ and $< \eta_S$ for finite I.

5 Demand Model: CGP and Sectoral Growth Rates

The Constant Growth Path (CGP)

On the **Constant Growth Path (CGP)**, aggregate consumption C_t grows at a constant rate g, but each sector's consumption grows at a *different* rate.

A CGP is a balanced-growth-like path on which:

1. Aggregate consumption grows at constant rate: $\dot{C}/C = g$.
2. Sectoral consumptions grow at (generally different) constant rates.
3. Relative prices remain constant.

Existence and uniqueness condition:

$$\boxed{\frac{\gamma^A}{B_A} = \frac{\gamma^S}{B_S}}$$

where B_i is a productivity normalization. This condition pins down the unique CGP.

Sectoral Consumption Growth Rates on the CGP

Let aggregate per-capita consumption be $X_t = B_M C_t$ (growing at rate g). Then:

$$\frac{\dot{c}_t^A}{c_t^A} = g \cdot \frac{c_t^A - \gamma^A}{c_t^A} < g \qquad \text{(Agriculture: grows \textit{slower} than } g\text{)}$$

$$\frac{\dot{c}_t^M}{c_t^M} = g \qquad \text{(Manufacturing: grows at } g\text{)}$$

$$\frac{\dot{c}_t^S}{c_t^S} = g \cdot \frac{c_t^S + \gamma^S}{c_t^S} > g \qquad \text{(Services: grows \textit{faster} than } g\text{)}$$

From the demand schedule $c_j = \gamma_j + \eta_j(I - \sum_i p_i\gamma_i)/p_j$, differentiate with respect to time (holding relative prices constant):

$$\dot{c}_j = \eta_j \dot{I}/p_j = \eta_j g \cdot X_t/p_j$$

So:

$$\frac{\dot{c}_j}{c_j} = \frac{\eta_j g X_t/p_j}{c_j} = g \cdot \frac{\eta_j X_t/p_j}{c_j} = g \cdot \frac{c_j - \gamma_j}{c_j}$$

- Agriculture ($\gamma^A > 0$): $(c^A - \gamma^A)/c^A < 1 \Rightarrow$ grows below g.
- Manufacturing ($\gamma^M = 0$): $(c^M - 0)/c^M = 1 \Rightarrow$ grows at g.
- Services ($\gamma^S < 0$): $(c^S + |\gamma^S|)/c^S > 1 \Rightarrow$ grows above g.

As $t \to \infty$, all sectoral growth rates converge to g (since $\gamma_i/c_i \to 0$).

On the CGP: agriculture grows below aggregate, manufacturing at aggregate, services above aggregate. The **agriculture share** in total consumption falls; the **services share** rises; manufacturing share is approximately stable. This matches the demand-side stylized facts without requiring any relative price changes.

6 Employment Dynamics under Demand-Driven Structural Change

Setup

Each sector i employs L_{it} workers and produces output $Y_{it} = B_i L_{it} F(k^*, 1) \cdot X_t$ (Cobb-Douglas, on balanced growth path with optimal capital per worker k^*). Market clearing in sector i:

$$L_{it} \cdot B_i \cdot F(k^*, 1) \cdot X_t = N_t \cdot c_t^i$$

where N_t is population and X_t is the aggregate productivity index.

Sectoral Employment Growth Rates

Differentiating the market clearing condition:

$$\frac{\dot{L}_t^A}{L_t^A} = n - \frac{g\,\gamma^A L_t}{B_A L_{At} X_t F(k^*, 1)} < n \qquad \text{(Agriculture)}$$

$$\frac{\dot{L}_t^M}{L_t^M} = n \qquad \text{(Manufacturing)}$$

$$\frac{\dot{L}_t^S}{L_t^S} = n + \frac{g\,\gamma^S L_t}{B_S L_{St} X_t F(k^*, 1)} > n \qquad \text{(Services)}$$

Interpretation

- Agricultural employment grows *slower* than population: the agriculture employment share L^A/L falls over time.
- Manufacturing employment grows *at* the population rate: the manufacturing share is approximately stable (in the long run converges to η_M).
- Services employment grows *faster* than population: the services employment share L^S/L rises over time.
- In the long run, as $c^i \gg \gamma^i$, all employment shares stabilize at η_i values.

Prediction for the Agriculture Employment Share

The agriculture employment share $l_t^A = L_t^A/L_t$ satisfies:

$$\dot{l}_t^A / l_t^A = \frac{\dot{L}_t^A}{L_t^A} - n = -\frac{g\,\gamma^A L_t}{B_A L_{At} X_t F(k^*, 1)} < 0$$

So l_t^A is monotonically decreasing. This matches the key stylized fact.

7 Supply-Side: Baumol's Cost Disease Framework

Two-Sector CES Economy

Consider a two-sector model (can be thought of as manufacturing vs services, or labor-intensive vs capital-intensive). The final good is a CES aggregate:

$$Y_t = \left[\omega\, Y_{1t}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega)\, Y_{2t}^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}} \qquad \text{(CES Aggregator)}$$

where $\varepsilon > 0$ is the elasticity of substitution, $\omega \in (0,1)$ is the share parameter.

Sectoral Production

$$Y_{it} = A_{it}\, K_{it}^{\alpha_i}\, L_{it}^{1-\alpha_i}, \qquad i = 1, 2 \qquad \text{(Sector Production)}$$

with sector-specific TFP growth:

$$\frac{\dot{A}_{it}}{A_{it}} = a_i > 0$$

Key asymmetry assumption: $\alpha_1 < \alpha_2$

Sector Asymmetry

Sector 1 is **more labor-intensive** ($\alpha_1 < \alpha_2$); Sector 2 is **more capital-intensive**. This capital-intensity difference is what generates factor reallocation dynamics.

Cases for Elasticity of Substitution

Case	Parameter Value	Relationship between goods
Complements	$\varepsilon < 1$	Hard to substitute; "need both"
Unit elastic (Cobb-Douglas)	$\varepsilon = 1$	Cobb-Douglas aggregator
Substitutes	$\varepsilon > 1$	Easy to substitute one for the other
Leontief	$\varepsilon \to 0$	Perfect complements; fixed proportions

8 Supply Model: Capital and Labor Allocation

Capital Allocation

The fraction of total capital allocated to sector 1:

$$k_t := \frac{K_{1t}}{K_t} \tag{Capital Share}$$

From the profit-maximizing firm's FOC equating marginal products of capital across sectors:

$$k_t = \left[1 + \frac{(1-\omega)}{\omega} \cdot \frac{\alpha_2}{\alpha_1} \cdot \left(\frac{Y_{2t}}{Y_{1t}}\right)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{-1}$$

Labor Allocation

The fraction of total labor in sector 1:

$$\lambda_t := \frac{L_{1t}}{L_t} \tag{Labor Share}$$

From equating marginal products of labor:

$$\lambda_t = \left[1 + \frac{(1-\alpha_2)}{(1-\alpha_1)} \cdot \frac{\alpha_1}{\alpha_2} \cdot \frac{1-k_t}{k_t}\right]^{-1}$$

Key relationship: λ_t is **increasing in** k_t. Labor and capital move in the *same direction*. If capital flows to sector 1, labor also flows to sector 1.

With Cobb-Douglas production in each sector and competitive factor markets, the ratio of labor to capital in each sector is determined by the wage-rental ratio w/r, which is common across sectors. A higher k_t (more capital in sector 1) means sector 1 has absorbed more productive inputs, raising its demand for labor too. The elasticity relationship $\partial\lambda_t/\partial k_t > 0$ follows from the cross-derivative of the profit function.

TFP-Driven Reallocation

The direction of TFP-driven factor flows:

$$\frac{d\ln k_t}{d\ln A_{2t}} = -\frac{d\ln k_t}{d\ln A_{1t}} > 0 \quad \Longleftrightarrow \quad \varepsilon < 1 \tag{TFP Channel}$$

Interpretation: when $\varepsilon < 1$, a TFP increase in sector 2 *draws capital away from sector 2 and toward sector 1*. Counterintuitive, but rational: the fast-growing sector needs fewer inputs to produce its required output.

9 Key Result: Reallocation Direction ($\varepsilon < 1$ vs $\varepsilon > 1$)

Main Theorem

$$\frac{d \ln k_t}{d \ln K_t} > 0 \quad \Longleftrightarrow \quad (\alpha_2 - \alpha_1)(1 - \varepsilon) > 0$$

Since $\alpha_2 > \alpha_1$ by assumption, this simplifies to:

$$\frac{d \ln k_t}{d \ln K_t} > 0 \quad \Longleftrightarrow \quad \varepsilon < 1 \quad \text{(goods are complements)}$$

Two Cases

	Complements ($\varepsilon < 1$)	**Substitutes** ($\varepsilon > 1$)
Capital reallocation	Capital flows toward sector 1 (labor-intensive) as K grows	Capital flows toward sector 2 (capital-intensive) as K grows
Matching stylized fact	Matches: labor shifts from labor-intensive agriculture to capital-intensive manufacturing	Does not match observed reallocation
Intuition	Need both goods; producing more of the scarce good requires the labor-intensive sector to grow	Substitute away from expensive-factor good; capital stays where it has comparative advantage
Factor flows	Capital and labor move together toward sector 1	Capital and labor stay in sector 2

Baumol's Cost Disease: Key Equation

From the growth-rate accounting of the capital allocation:

$$\gamma_{K1} - \gamma_{K2} = \frac{\varepsilon - 1}{\varepsilon} (\gamma_{Y1} - \gamma_{Y2}) \qquad \text{(Baumol)}$$

Since labor follows capital (λ_t is increasing in k_t), the same sign applies to labor:

If $\varepsilon < 1$ (complements):

$$\gamma_{L1} > \gamma_{L2} \iff \gamma_{K1} > \gamma_{K2} \iff \gamma_{Y1} < \gamma_{Y2}$$

The **slow-growing sector absorbs factors**. Reallocation goes *opposite* to sectoral productivity growth.

If $\varepsilon > 1$ (substitutes):

$$\gamma_{L1} < \gamma_{L2} \iff \gamma_{K1} < \gamma_{K2} \iff \gamma_{Y1} < \gamma_{Y2}$$

The **fast-growing sector absorbs factors**. Reallocation goes *in the same direction* as productivity growth.

Baumol's Cost Disease Interpretation

Services have *low* TFP growth. If services and manufacturing are complements ($\varepsilon < 1$):

- Services become relatively expensive (price rises relative to manufacturing).
- But demand is inelastic, so nominal spending on services rises.
- Factors (labor) flow *into* services, even though services are becoming more expensive.
- This is the "cost disease": rising costs in services do not reduce demand due to complementarity.

Key Insight to Memorize

The Baumol insight: When $\varepsilon < 1$, the low-TFP-growth sector (services) absorbs factors over time, even as it becomes relatively more expensive. This explains the services employment share rising monotonically.

10 Testing Demand vs Supply: Herrendorf, Rogerson & Valentinyi (2013)

Why the Test Matters

Both demand and supply mechanisms can in principle generate the observed structural change patterns. Herrendorf, Rogerson & Valentinyi (HRV) ask: which mechanism quantitatively dominates, and can we identify the structural parameters?

HRV Preferences

HRV use the following non-homothetic CES preferences:

$$u_t = \left[\sum_{i \in \{a,m,s\}} \omega_i^{1/\sigma} \left(c_{it} - \bar{c}_i\right)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \tag{HRV Utility}$$

where:

- $\sigma > 0$: intertemporal/intratemporal elasticity of substitution.
- ω_i: weight on sector i in the aggregator.
- $\bar{c}_i$: non-homothetic threshold (positive $\Rightarrow$ necessity; negative $\Rightarrow$ luxury).

Two Approaches to Measurement

	Final Expenditure Approach	**Value Added Approach**
What is measured	What households actually buy (e.g., food in restaurants, movies, cars)	Value added in each sector's production (agriculture, manufacturing, services as classified by GDP accounts)
What substitution is	Substituting between consumer goods (e.g., eating out vs grocery shopping)	Substituting between bundles of factors organized by industry

HRV Empirical Results

Parameter	**Final Expenditure**	**Value Added**
σ (elasticity of substitution)	0.85	≈ 0.002 (nearly Leontief!)
$\bar{c}_a$ (agriculture threshold)	-1350.38 (luxury!)	-138.68
$\bar{c}_s$ (services threshold)	11,237.40 (necessity)	4,261.82
ω_s (services weight)	0.81	—

Interpretation of the Divergence

Final expenditure: When a consumer chooses between watching a movie (services) and attending a football match (also services), or between a restaurant meal and a supermarket meal, these are *genuine substitutes* from the consumer's perspective. Hence $\sigma \approx 0.85$: moderate substitutability.
Value added: A restaurant meal and supermarket food use *very similar factor inputs* (labor, capital). In GDP accounts, a restaurant meal is classified as *services* but supermarket food as *agriculture*. Since their factor compositions are almost identical, the value-added aggregator has essentially Leontief complementarity: $\sigma \approx 0.002$.
Implication: The "structural change" seen in value-added sector shares partly reflects how national accounts classify goods, not genuine consumer substitution. This matters for identifying the demand vs supply mechanism.

Classification Puzzle: Agriculture as a Luxury

A striking HRV result is that agriculture has $\bar{c}_a < 0$ in the final expenditure approach, making it a **luxury**. This seems to contradict Engel's Law. Why?

In the **final expenditure** classification, what counts as "agriculture" includes restaurant meals, prepared foods, premium ingredients — goods that are luxuries. Low-income consumers eat cheap staples (classified as agriculture); high-income consumers eat processed foods and restaurant meals (classified as services). So in final expenditure terms, *agriculture spending* can rise more than proportionally with income.
In the **value added** classification, agriculture's $\bar{c}_a < 0$ still holds but is less extreme, because the distinction is by production process (farming vs manufacturing vs services).

11 Empirical Results and Literature

Summary of Key Papers

Paper	Mechanism / Contribution	Key Finding / Limitation
Duarte & Restuccia (2010)	Documents TFP growth differentials across sectors 1956–2004	Agriculture 3.8%, Manufacturing 2.4%, Services 1.3%; TFP gaps explain much of cross-country income differences
Herrendorf, Rogerson & Valentinyi (2013)	Tests demand vs supply using two measurement approaches	Final expenditure: $\sigma \approx 0.85$, demand matters. Value added: $\sigma \approx 0$, supply (Baumol) dominates; reconciles the two views
Buera & Kaboski (2009)	Stone-Geary + sector-biased technical change	Cannot match the steep manufacturing employment decline at high income levels
Boppart (2014)	PIGL (Price-Independent Generalized Linear) preferences	Non-homotheticities do not vanish asymptotically; demand and supply each explain roughly 50% of structural change
Comin, Lashkari & Mestieri (2021)	Non-homothetic CES preferences	R^2 twice as large as simpler models; non-homotheticities explain approximately 3/4 of structural change

Demand vs Supply: Side-by-Side Comparison

Feature	Demand (Non-Homotheticity)	Supply (Baumol Cost Disease)
Key mechanism	Income effects: as income rises, budget shares shift toward luxuries (services) and away from necessities (agriculture)	Price effects: productivity differences change relative prices; with $\varepsilon < 1$, factors flow toward slow-TFP sector
Key parameter	Income elasticities ($\bar{c}_i$ in Stone-Geary / HRV)	Elasticity of substitution ε
Relative prices	Structural change occurs even with *constant* relative prices	Relative price changes drive factor reallocation
Cross-country variation	Predicts SC correlated with per-capita income level	Predicts SC driven by TFP growth differentials
Long-run behavior	Non-homotheticities vanish as $I \to \infty$ (sector shares converge to η_i)	Persists as long as TFP growth rates differ
Evidence	HRV: $\sigma \approx 0.85$ (final expenditure); non-homothetic CES fits well	HRV: $\sigma \approx 0$ (value added); Baumol mechanism important
Limitation	Cannot explain all of manufacturing's inverse-U pattern	With $\varepsilon > 1$, predicts wrong direction of reallocation
Best suited for	Cross-section: poor vs rich countries	Time-series: explaining secular increase in services share

12 Problem-Solving Templates

(a) Derive Stone-Geary Demand Schedules

1. Write the Stone-Geary utility: $u = \prod_i (c_i - \gamma_i)^{\eta_i}$.
2. Set up the Lagrangian with budget constraint $\sum_i p_i c_i = I$.
3. Derive FOC for good j: $\eta_j u/(c_j - \gamma_j) = \lambda p_j$.
4. Take ratios of FOCs to eliminate λ: $\eta_i(c_j - \gamma_j)p_j = \eta_j(c_i - \gamma_i)p_i$.
5. Substitute into budget constraint. Use $\sum_i \eta_i = 1$ to simplify.
6. Result: $c_j = \gamma_j + \eta_j(I - \sum_i p_i\gamma_i)/p_j$.
7. Verify: check that $\sum_i p_i c_i = I$ holds as a consistency check.

(b) Compute Engel Curve Slope and Income Elasticity

1. From demand $c_j = \gamma_j + \eta_j(I - \sum_k p_k\gamma_k)/p_j$, differentiate w.r.t. I:
$$\frac{\partial c_j}{\partial I} = \frac{\eta_j}{p_j} > 0 \quad \text{(always a normal good)}$$
2. Income elasticity:
$$\varepsilon_I^j = \frac{\partial c_j}{\partial I} \cdot \frac{I}{c_j} = \frac{\eta_j I}{p_j c_j}$$
3. Substitute $p_j c_j = p_j\gamma_j + \eta_j(I - \sum_k p_k\gamma_k)$:
$$\varepsilon_I^j = \frac{\eta_j I}{p_j\gamma_j + \eta_j(I - \sum_k p_k\gamma_k)}$$
4. Sign of $\varepsilon_I^j - 1$: positive (luxury) iff $p_j\gamma_j < 0$ (i.e., $\gamma_j < 0$); negative (necessity) iff $\gamma_j > 0$.
5. As $I \to \infty$: $\varepsilon_I^j \to \eta_j < 1$ for agriculture (since $\gamma^A > 0$ is finite and I dominates).

(c) Derive CGP Growth Rates

1. Start from demand: $c_j = \gamma_j + \eta_j(I - \sum_k p_k\gamma_k)/p_j$.
2. Differentiate with respect to time (holding prices constant): $\dot{c}_j = \eta_j \dot{I}/p_j$.
3. Identify $\dot{I}/p_j$ in terms of $c_j - \gamma_j$: from demand, $(c_j - \gamma_j) = \eta_j(I - \sum p_k\gamma_k)/p_j$, so $\eta_j\dot{I}/p_j = \dot{I}/(I - \sum p_k\gamma_k) \cdot (c_j - \gamma_j)$.
4. Define $g = \dot{I}/(I - \sum p_k\gamma_k)$ as the aggregate growth rate of supernumerary income.
5. Sector growth rate: $\dot{c}_j/c_j = g(c_j - \gamma_j)/c_j$.
6. For agriculture ($\gamma^A > 0$): ratio $< 1 \Rightarrow$ grows below g.
7. For services ($\gamma^S < 0$): ratio $> 1 \Rightarrow$ grows above g.

(d) Apply the Baumol Reallocation Condition

1. Identify which sector is more capital-intensive: $\alpha_2 > \alpha_1$ (sector 2 capital-intensive).
2. Identify elasticity of substitution ε from the problem.
3. Apply the sign rule:
   - $\varepsilon < 1$: $d \ln k_t / d \ln K_t > 0 \Rightarrow$ capital flows toward sector 1 (labor-intensive).
   - $\varepsilon > 1$: $d \ln k_t / d \ln K_t < 0 \Rightarrow$ capital flows toward sector 2 (capital-intensive).
4. Use the Baumol equation: $\gamma_{K1} - \gamma_{K2} = [(\varepsilon - 1)/\varepsilon](\gamma_{Y1} - \gamma_{Y2})$.
5. Since λ_t is increasing in k_t: labor and capital flow in the same direction.
6. Conclude on the direction of labor reallocation and which sector's employment share rises.

(e) Identify Demand vs Supply Channel (Empirical Strategy)

1. Compare value-added vs final expenditure measures of sector shares and relative prices.
2. Estimate σ / ε from each approach separately.
3. **If σ (final expenditure) is substantially higher than σ (value added)**: evidence that classification matters and supply mechanism is important in value added.
4. Check $\bar{c}_i$ signs: negative $\bar{c}_a$ in final expenditure $\Rightarrow$ agriculture classified as luxury in consumer spending (HRV finding).
5. Use $\sigma \approx 0$ (value added) to argue Baumol/supply dominates in national accounts.
6. Use $\sigma \approx 0.85$ (final expenditure) to argue demand substitution matters for consumer allocation.

13 Common Pitfalls

1. **Confusing homotheticity with unit income elasticity everywhere**
   - Homothetic preferences have income elasticity of 1 for all goods, so all Engel curves pass through the origin.
   - Stone-Geary is *not* homothetic: income elasticities differ across goods and depend on I.
   - Do not say "Stone-Geary is homothetic because it has linear Engel curves": linearity without passing through origin is not homotheticity.
2. **Forgetting the sign of γ^S in Stone-Geary**
   - The standard formulation is $u = (c^A - \gamma^A)^{\eta_A}(c^M)^{\eta_M}(c^S + \gamma^S)^{\eta_S}$ with $\gamma^A, \gamma^S > 0$.
   - The $+\gamma^S$ means services acts as a luxury (negative threshold in the unified notation $c_i - \gamma_i$ with $\gamma^S_{\text{unified}} < 0$).
   - Always verify: $c^A - \gamma^A > 0$ must hold (consumption above subsistence).
3. **Misidentifying the direction of Baumol reallocation**
   - The key result is: with $\varepsilon < 1$, factors flow toward the *slow-TFP-growth* sector.
   - Many students incorrectly say "factors follow productivity". The opposite is true under complementarity.
   - Mnemonic: "slow sector starves for inputs because you cannot substitute away from it".
4. **Applying $\varepsilon > 1$ intuition when the model uses $\varepsilon < 1$**
   - For structural change to match stylized facts in the supply model, we need $\varepsilon < 1$.
   - Do not apply standard comparative advantage arguments (factors flow to productive sector) — that reasoning applies when $\varepsilon > 1$ only.
5. **Confusing the CGP with a balanced growth path**
   - On a BGP, all variables grow at the same rate. On the CGP, sector growth rates differ.
   - The CGP is the correct concept for structural change: it allows sector heterogeneity while keeping aggregate growth constant.
   - Existence of the CGP requires the specific condition $\gamma^A/B_A = \gamma^S/B_S$.
6. **Conflating final expenditure and value added in empirical analysis**
   - HRV's key insight is that final expenditure and value added give very different σ estimates.
   - Do not assume they should agree; their divergence reveals information about the classification of goods.
   - When told "services and manufacturing are complements", check whether this refers to final goods or factor-input bundles.
7. **Claiming structural change requires relative price changes**
   - With demand non-homotheticities (Stone-Geary), structural change occurs even at *constant relative prices.*

- Income growth alone shifts budget shares if income elasticities differ.
- Relative price changes are necessary only for the supply (Baumol) mechanism.

8. **Ignoring that labor and capital move together**

   - In the supply model, λ_t is increasing in k_t, so labor always follows capital.
   - Once you determine which sector receives more capital, the labor allocation is determined.

14 Quick Reference Formulas

Stone-Geary Preferences

$$u = (c^A - \gamma^A)^{\eta_A}(c^M)^{\eta_M}(c^S + \gamma^S)^{\eta_S}, \quad \eta_A + \eta_M + \eta_S = 1 \tag{Utility}$$

$$c_j = \gamma_j + \frac{\eta_j(I - \sum_i p_i\gamma_i)}{p_j} \tag{Demand}$$

$$\frac{\partial c_j}{\partial I} = \frac{\eta_j}{p_j}, \qquad \varepsilon_I^j = \frac{\eta_j I}{p_j c_j} \tag{Engel Curve}$$

CGP Growth Rates

$$\frac{\dot{c}^A}{c^A} = g \cdot \frac{c^A - \gamma^A}{c^A} < g \tag{Agriculture}$$

$$\frac{\dot{c}^M}{c^M} = g \tag{Manufacturing}$$

$$\frac{\dot{c}^S}{c^S} = g \cdot \frac{c^S + \gamma^S}{c^S} > g \tag{Services}$$

Employment Growth Rates (Demand Model)

$$\frac{\dot{L}^A}{L^A} = n - \frac{g\,\gamma^A L}{B_A L^A X F(k^*, 1)} < n \tag{Agriculture}$$

$$\frac{\dot{L}^M}{L^M} = n \tag{Manufacturing}$$

$$\frac{\dot{L}^S}{L^S} = n + \frac{g\,\gamma^S L}{B_S L^S X F(k^*, 1)} > n \tag{Services}$$

Supply Model: CES Aggregator and Allocation

$$Y_t = \left[\omega Y_{1t}^{\frac{\varepsilon-1}{\varepsilon}} + (1-\omega) Y_{2t}^{\frac{\varepsilon-1}{\varepsilon}}\right]^{\frac{\varepsilon}{\varepsilon-1}} \tag{CES}$$

$$k_t = \left[1 + \frac{1-\omega}{\omega} \cdot \frac{\alpha_2}{\alpha_1} \cdot \left(\frac{Y_{2t}}{Y_{1t}}\right)^{\frac{\varepsilon-1}{\varepsilon}}\right]^{-1} \tag{Capital Allocation}$$

$$\lambda_t = \left[1 + \frac{1-\alpha_2}{1-\alpha_1} \cdot \frac{\alpha_1}{\alpha_2} \cdot \frac{1-k_t}{k_t}\right]^{-1}, \quad \frac{d\lambda_t}{dk_t} > 0 \tag{Labor Allocation}$$

Key Reallocation Results

$$\frac{d\ln k_t}{d\ln K_t} > 0 \iff (\alpha_2 - \alpha_1)(1-\varepsilon) > 0 \iff \varepsilon < 1 \tag{Capital Direction}$$

$$\gamma_{K1} - \gamma_{K2} = \frac{\varepsilon - 1}{\varepsilon}(\gamma_{Y1} - \gamma_{Y2}) \tag{Baumol}$$

$$\frac{d\ln k_t}{d\ln A_{2t}} > 0 \iff \varepsilon < 1 \tag{TFP Channel}$$

HRV Utility and Empirical Estimates

$$u_t = \left[\sum_i \omega_i^{1/\sigma} (c_{it} - \bar{c}_i)^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} \tag{HRV}$$

Parameter	Final Expenditure	Value Added	Implication
σ	0.85	0.002	Leontief in VA; substitutable in FE
$\bar{c}_a$	-1350.38	-138.68	Agriculture is luxury in FE
$\bar{c}_s$	11,237.40	4,261.82	Services is necessity

Classification of Goods (Quick Reference)

Sector	Income Elasticity	γ_i Sign (Stone-Geary)
Agriculture	< 1 (necessity; $\varepsilon_I^A \to \eta_A$ as $I \to \infty$)	$\gamma^A > 0$
Manufacturing	≈ 1	$\gamma^M = 0$
Services	> 1 for moderate I (luxury)	$\gamma^S < 0$ (i.e., $+\gamma^S$ in utility with $\gamma^S > 0$)